

Lecture 26

↳ Shor's 9 qubit code

↳ can correct any type of single-qubit error

$$-X, Y, Z, I, X+iY, e^{i\theta X}, \dots$$

↳ Fault Tolerance

→ ability to correct errors using faulty error correction circuits

→ classical circuit example.

3 qubit bit flip code:

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle$$

$$X_L = X \otimes X \otimes X$$

$$Z_L = I \otimes I \otimes Z_0$$

or Z_1
 Z_2

you have to flip all 3 bits to go from 1 logical state to the other

⇒ Code distance for bit flips = 3.

⇒ code distance for phase flips = 1.

Introduce stabilizers to detect bit flip errors:

$$\begin{cases} S_1 = Z_2 Z_1 \\ S_0 = Z_1 Z_0 \end{cases}$$

Phase flip code

$$|0_L\rangle = |+++ \rangle$$

$$|1_L\rangle = |-- \rightarrow$$

Shor construction: start with phase flip code

$$|0_L\rangle = |+++ \rangle = |+\rangle^{\otimes 3}$$

$$|1_L\rangle = |--\rangle = |-\rangle^{\otimes 2}$$

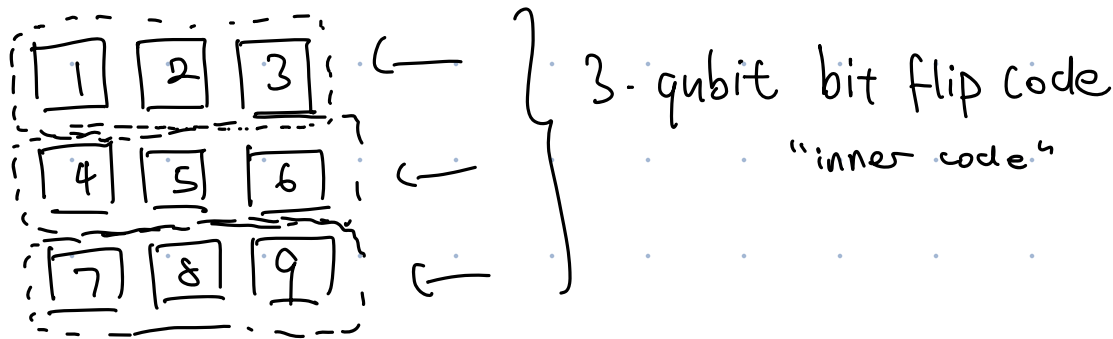
Then encode bit flip.

$$|+\rangle \rightarrow \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)$$

$$|0_L\rangle \rightarrow \frac{1}{\sqrt{8}} (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle)$$

they are entangled

$$|1_L\rangle \rightarrow \frac{1}{\sqrt{8}} (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle)$$



$$X_L = (Z_1 Z_2 Z_3) (Z_4 Z_5 Z_6) (Z_7 Z_8 Z_9) \nearrow \text{distance } 3$$

$$= Z_1 \otimes Z_4 \otimes Z_7$$

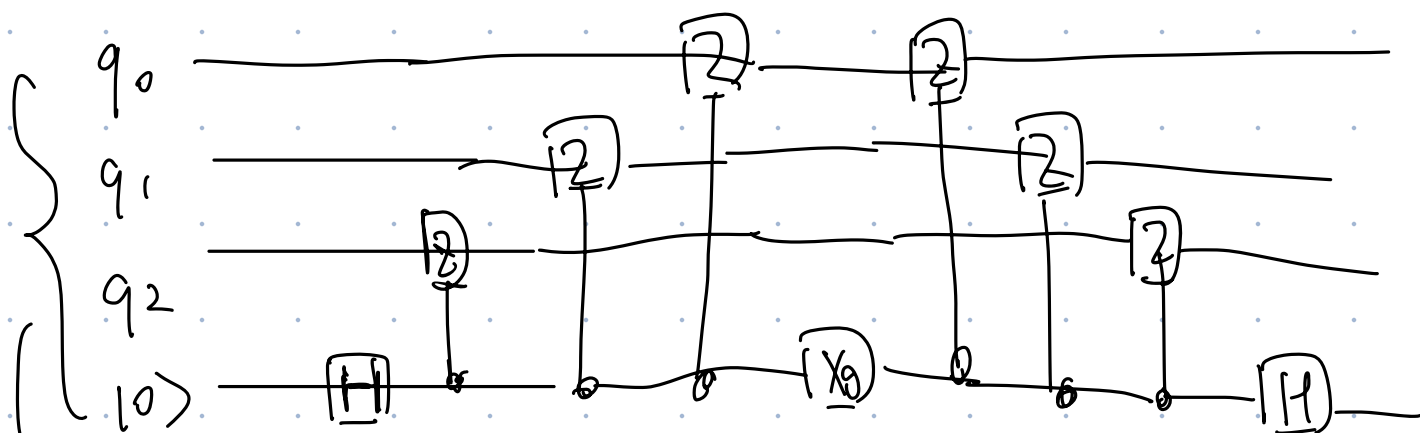
↑
You only need to apply 3 Z's
to turn a $|0_L\rangle \rightarrow |1_L\rangle$

$$Z_L = (X_1 X_2 X_3) (X_4 X_5 X_6) (X_7 X_8 X_9)$$

$$= \underbrace{X_1 X_2 X_3}_{\text{only } \pi/4}$$

Daniel
Weissman?
PhD

$$e^{-i\frac{\theta}{2} Z \otimes Z \otimes Z} = \cos\frac{\theta}{2} I \otimes I \otimes I - i \sin\frac{\theta}{2} Z \otimes Z \otimes Z$$



makes it hard for me to do a logical gate!
 makes it hard for errors to affect information

9 physical qubits $\rightarrow$ one logical qubit

$$[n, k, d] = [9, 1, 3]$$

classical code 1001000

99,000

in cell phone.

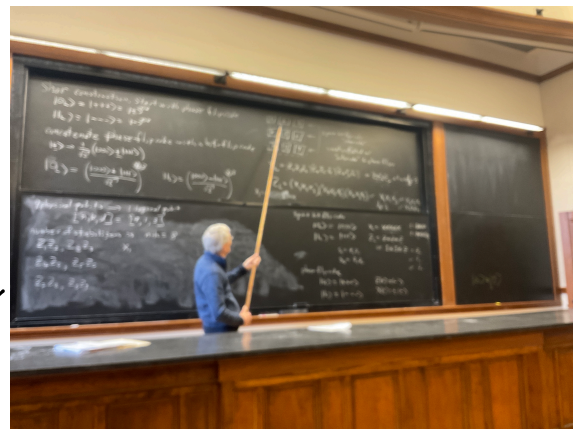
9 qubits

$$\text{Number of stabilizers} = n - k = 8$$

↓ $\Rightarrow$ Abelian Group

$z_1 z_2, z_2 z_3$
 $z_4 z_5, z_5 z_6$
 $z_7 z_8, z_8 z_9$

} stabilizer generators



$$(X_1 X_2 X_3)(X_4 X_5 X_6)$$

$$(X_4 X_5 X_6)(X_7 X_8 X_9)$$

PHYS error set

$I, X_1, \dots, X_9, Y_1, \dots, Y_9, Z_1, \dots, Z_9 \Rightarrow 28$ errors at minimum.

Example: 1 Error on S, X_5 :

is just see which one does not commute on X_5 .

$$\begin{aligned}
 X_5 |4\rangle &= X_5 S_j |4\rangle \\
 &= \pm S_j [X_5 |4\rangle]
 \end{aligned}$$

"-" eigenvalue: S_j known there's an error

Does not commute with

$$z_4 z_5, z_5 z_6$$

$$\downarrow$$

$$-1$$

$$\downarrow$$

$$-1$$

identity on X_5 error



apply X_5 .

error Z_5 anticommutes $(X_1 X_2 X_3) (X_4 X_5 X_6)$

$$(X_4 X_5 X_6) (X_1 X_2 X_3)$$

this is not unique.

$$\text{apply } Z_4(Z_5|4\rangle) = (Z_4 Z_5)|4\rangle$$

$\bigcirc \bigcirc \bigcirc$

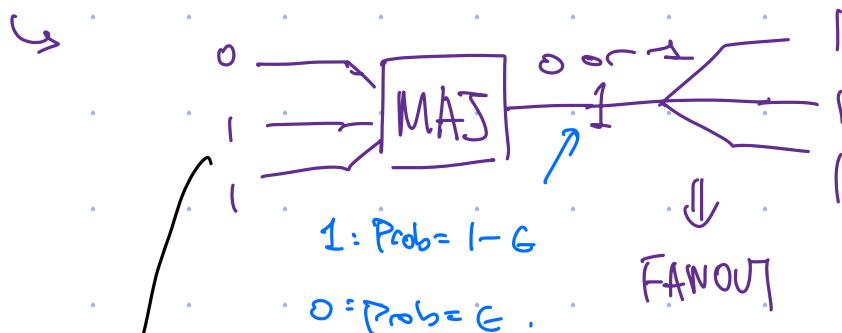
this is a stabilizer on a no-error state.

should apply

$Z_5(Z_5|4\rangle)$ to correct the error

Fault-Tolerance

↳ 3-classical bit repetition code
(000) (111)



what can you do

